

LumenArc: Evidence-First Crypto Research Across Thirty Reports

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Problem and intended user

A digital-asset research analyst must answer precise questions from long research reports while market conditions and news continue to change. A plausible but invented resistance level is worse than an honest refusal, especially when a reader might mistake the answer for a trading signal. LumenArc turns original research documents into cited answers, an evidence-linked brief, and a separate live context dashboard. The core job is to find the number or sentiment that a named report actually states, show its page and quotation, or say that the source is silent. Trade execution, personalized investment advice, and price prediction are explicitly outside scope.

The initial commitment was to analyze exactly thirty PDF reports and compare against a keyword baseline. That commitment is met with thirty original [Binance Research monthly reports](#), not thirty invented entries. Their source URLs, SHA256 checksums, titles, access results, and page counts are recorded. The original PDFs were successfully fetched and processed into 559 PDF pages and 1,017 searchable passages. Their complete bytes are not redistributed because public access does not establish republication rights. A documented importer retrieves them from the publisher.

Implementation and technical choices

A React and TypeScript client provides a bilingual English and Chinese research workspace. A Node and Express backend validates and parses genuine PDF, DOCX, text, and Markdown files, retains PDF page boundaries, ranks passages lexically, and owns every external connection and protected AI request. For document questions, the rented model receives only retrieved excerpts and must return a structured answer with passage identifiers. A deterministic check rejects unsupported newly asserted numbers and missing citations. The interface uses categorical evidence states, not an uncalibrated confidence percentage. A separately selectable keyword baseline returns its best matching sentence without model interpretation. A brief synthesizes only cited source material.

Streamlit was originally attractive because it could produce a first interface quickly. React and Node were deliberately chosen instead. They give a clearer frontend and backend secret boundary, explicit document-workspace state, robust real-time connection health, and a more mature animated bilingual interface. This gained control over interaction and provenance at the

cost of more code, deployment complexity, and tests. The product owns source parsing, retrieval, refusal checks, and evaluation, while it rents a foundation model and outside feeds. This is a build-versus-buy decision rather than an unexplained change of framework.

The original Coinbase WebSocket was blocked by TLS certificate validation in this environment. Instead of disabling certificate checks or fabricating prices, the app uses the documented Binance Spot market-only WebSocket. Ten major assets are quoted against USDT, explicitly labeled as such rather than presented as exact fiat USD. Trade events drive price changes, ticker events supply rolling 24-hour changes, and connection, partial-feed, and stale states expose failures. One six-second check received 65 backend-relayed events across all ten assets. An on-demand AI market pulse interprets only a cited, timestamped observation of those prices, never a forecast. News comes from recent publisher RSS feeds and on-demand Google News searches. Optional AI context is limited to the returned headline and excerpt, not an unseen article body. News discovery refreshes periodically and is not represented as exchange-speed streaming.

Evaluation design and measured outcome

Twelve development questions from March to August 2024 selected retrieval and verification behavior. The fifty-question holdout uses different report months, with forty answerable numeric questions and ten no-answer questions. The holdout labels and configuration were checksummed before formal testing. The baseline and final system saw the same thirty parsed reports and questions. A numeric result counts only when the answer contains the expected normalized number and a cited passage in the correct original report contains it. A different valid page in the same report is accepted. This is stricter than matching a number alone, but it is not a proof of full semantic correctness.

Frozen holdout metric	Keyword baseline	Final AI system
Source-backed numeric answers, denominator 40	28, or 70.0 percent	37, or 92.5 percent
Correct refusals on unanswerable cases, denominator 10	0, or 0 percent	10, or 100 percent
Wrongful refusals on answerable cases, denominator 40	2, or 5.0 percent	3, or 7.5 percent
Unsupported responses on unanswerable cases, denominator 10	10, or 100 percent	0, or 0 percent

The final system improved source-backed numeric accuracy by 22.5 percentage points and refused all ten intended no-answer cases. It also wrongly withheld three answers that the baseline withheld only twice. Two failures were retrieval misses in the same monthly report. A third was marked Conflicting evidence because retrieved passages described both 20.2 percent and 13 percent declines for the queried period. The strict metric calls that a wrongful refusal even though flagging a source discrepancy may be prudent. Among forty answerable questions, thirty-four final answers cited the label's PDF page, while thirty-seven cited the correct report with the expected number. The raw predictions allow independent review.

The completed fifty-case run had a 3.55-second median and 4.99-second 90th-percentile per-case latency, including two immediate retrieval abstentions. Forty-eight cases recorded model token usage. The measured provider was the Manus project AI fallback, not the school OpenRouter key, because the protected secret input was not confirmed. The provider did not supply a reliable USD cost, so project AI spending is unknown rather than zero. The school allocation is a fixed US\$10 with no free quota or top-ups, and the application defaults to a bounded request count. No school-key expenditure is claimed. Two earlier incomplete runs failed on malformed model JSON, wrote no final summary, and remain logged. A documented format-only retry was added without altering the questions, retrieval, model, or answer rules.

Risks, limitations, and next steps

These fifty labels were authored from original passages and machine-checked against source-page quotations. They have not been independently human verified in full. A student should inspect every label before presenting the set as hand-verified. All thirty reports come from one publisher, so transfer to other document formats and editorial styles remains unproven. Phrase searches used for no-answer candidates also cannot prove that no paraphrase exists elsewhere in a report. Flattened tables, contradictory publisher statements, and semantically irrelevant nearby numbers can defeat automated evidence checks. An always-refusing system would score well on safe refusal while providing no useful answers, which is why wrongful refusal is reported separately.

Uploaded files reside temporarily in server memory, partitioned by a browser workspace identifier but without authentication or durable storage. Selected passages leave the server for the external AI provider when AI mode is used. Confidential material is inappropriate for this prototype. Exchange quotes are single-venue USDT prices, not a consolidated or guaranteed fiat rate. The system is not an autonomous investment agent. The initial 40 percent analyst-time saving claim was dropped because no analyst was timed against a stopwatch. Before production use, add account authentication, durable encrypted storage, independent human adjudication of labels, formal cost enforcement, and review of financial harm scenarios. The runnable code, product box diagram, import manifest, freeze snapshot, and [raw evaluation results](#) make both the achievement and its remaining uncertainty inspectable.